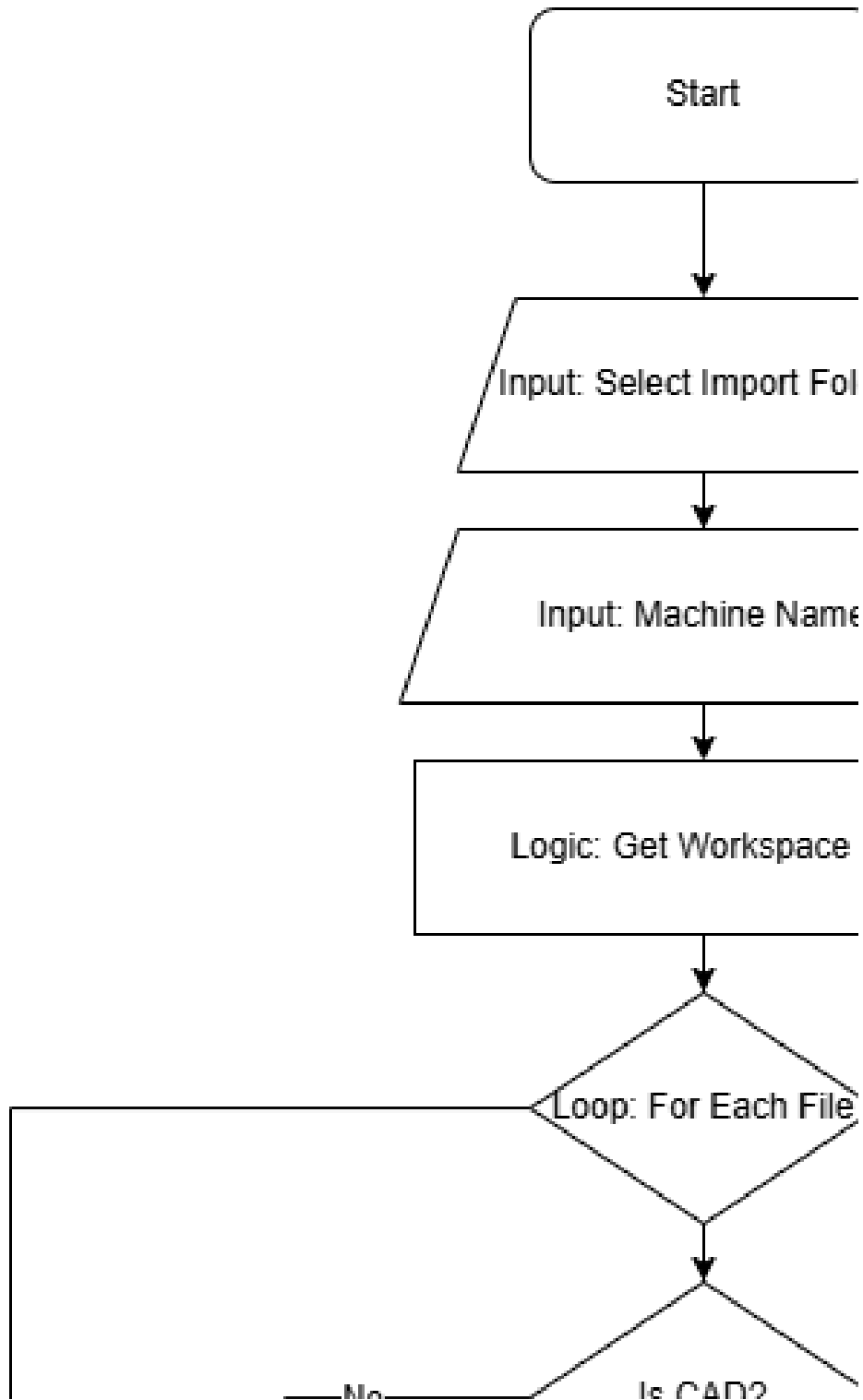


# Import File to Workspaces Script (MJ)

**Script Name:** import\_file\_to\_workspaces\_MJ.lua



# Functionality

This script automates the batch import of part files (both Mesh and CAD) from a directory. For each file, it creates a dedicated Workspace (tray) assigned to a specific machine, ensuring accurate build time and cost estimates.

## Logic Flow:

- **Machine Selection:** The user specifies a target machine (e.g., "Fuse 1"), and the script resolves the internal Workspace ID.
- **Batch Processing:** It iterates through every file in the selected directory.
- **Importer Selection:**
  - **Standard Meshes (STL, 3MF):** Imported directly.
  - **CAD Files (STEP):** Imported using a specialized loop that tries 11 internal kernel indices (0-10) to find a compatible importer. It also supports custom tessellation parameters.
- **Placement:** Each part is placed in its own new workspace and automatically centered.

## User Guide (Quick Start)

1. Prepare a folder containing all the files you wish to process.
2. Run the script **import\_file\_to\_workspaces\_MJ**.
3. Select the import folder when prompted.
4. Enter the exact **Machine Name** (as listed in "My Machines") when asked (default is "Formlabs Fuse 1").
5. If prompted for STEP parameters (Deviation, Angle, Edge Length), accept the defaults or customize them.
6. The script will generate a separate workspace for every file.

## Code Maintenance & Technical Notes

**Workspace vs. Tray:** The script explicitly uses `trayHandler:addworkspace(workspaceID)` instead of generic tray creation. This ensures that the specific machine configuration (and thus "No-Build" zones) is applied.

**STEP File Handling:** Standard mesh loaders fail on STEP files. The script implements a robust `loadcadfile` function that attempts to create a CAD importer using indices 0 through 10. This loop is critical for compatibility across different Netfabb versions and kernel states.

**Centering Logic:** Parts are centered based on the machine dimensions retrieved from the new workspace (``machinesize_x``, ``machinesize_y``) relative to the part's bounding box.